FraAlliance

Frankfurt Airport Operating Direction Forecast

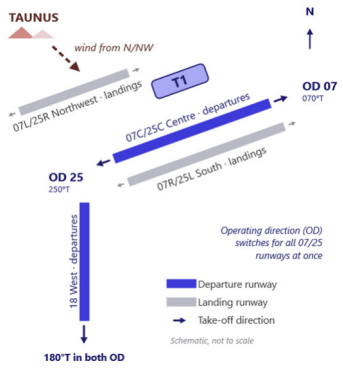
VENUE	DATE	FORMAT
AI Week 2026, Frankfurt	30. Oct 2026	One day · approx. 6 h build time

1 The challenge

At Frankfurt Airport, the wind mainly determines in which direction aircraft take off and land: shifting winds force an airport-wide change of operating direction, while strong northerly winds (e.g. from the **Taunus**) close Runway 18 (only used for departures). **Your mission:** build a **machine-learning model** that forecasts (1) **whether departures are possible on RWY 18** and (2) **which operating direction** is in use.

2 Runway system

- Frankfurt Airport has four runways: two (mainly) for **landings** (07L/25R Northwest, 07R/25L South) and two for **departures** (07C/25C Centre, 18 West).
- The 07/25 runways can be used in both directions, but the **operating direction (OD) is set airport-wide:** when it changes, **all runways switch at the same time**. Mixed directions are not possible.
- **OD 25:** departures and landings towards the west (250°T). **OD 07:** towards the east (070°T). **Runway 18 West:** departures to the south (180°T), used in both OD.



Schematic · T1 = Terminal 1

3 Decision rules

A · Wind direction and Tailwind component

Only the part blowing **exactly from behind** counts as tailwind. This **tailwind component must stay below 15 kt for departures and below 10 kt for landings.**

Crosswinds > 40 kts (incl. Gusts) force a runway closure.

- Example, 20 kt wind: 30° off straight behind → 17.3 kt → **not permitted**; 45° off → 14.1 kt → **permitted**.

B · Operating Direction (07/25)

- **Headwind is always preferred.**
- **Landings decide the OD: a tailwind component ≥ 10 kt on the landing runways forces an immediate change of OD.**
- **Persistence:** a **small tailwind** (always < 10 kt) **is accepted for short periods** (up to 30 min) **to avoid frequent changes;** if it persists longer, the OD switches to the headwind direction.
- Example: 090°/20 kt → ~19 kt tailwind on 25 → **OD 07**.

C · Runway 18

Departure Runway only, take-offs only towards the south (180°T),

- **Tailwind ≥ 15 kt** (wind from the north) → **no take-off on 18** means departures only via the Centre runway (07C/25C).
- Example: 330°/20 kt → 17,3 kt tailwind → Runway 18 closed.

4 Deliverables

1. **Machine-learning forecast model:** for every hour up to t+[48] h, predict the probability of RWY18 usability as well as the operating direction (25 / 07) with a certainty score and the next probable change time. Section 3 applied to a weather forecast is only the **baseline,** beat it.
2. **Self-sourced data:** no dataset is provided; each participant finds and combines suitable public data (e.g. METAR EDDF, weather-model forecasts, historical operating directions) and cites every source.
3. **Model quality:** show how good your model is - test it on data it was **not** trained on, using only data available at forecast time, and report the results for forecasts 1 h, 6 h and 24 h ahead.
4. **Submission at the end of the day:** code repository, one-page model card (data sources, model quality, key drivers of the forecast), forecasts for the jury test days. The forecasts are compared with the operating directions actually flown.